Personalized Recommendation with Contextual Bandits (NLP + RL) – Recommender System

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ABSTRACT

This paper presents a personalized news recommendation system combining NLP text embeddings with contextual bandit reinforcement learning (LinUCB). The system processes the MIND (Microsoft News Dataset) using hashed bag-of-words text vectors, categorical hashing, and pre-trained entity embeddings. Four baselines — random, popularity-based, content-similarity, and a lightweight MLP — are evaluated against LinUCB, which adaptively learns user preferences via online exploration-exploitation. Offline evaluation demonstrates that LinUCB outperforms all baselines on click-through rate and average reward. Key contributions include a reproducible offline pipeline, ethical risk assessment, and fairness slice analysis across content categories.

1. PROBLEM & MOTIVATION

Online news platforms face exponential growth in content volume, making personalised ranking critical for user engagement and discovery. Traditional recommendation systems suffer from three core pathologies:

- Popularity bias - algorithmic amplification of already-trending content suppresses long-tail articles.
- Filter bubbles - over-personalisation narrows user perspectives over time.
- Cold-start - systems cannot serve new users or items with limited interaction history.

This project addresses these challenges via a contextual bandit approach that balances exploration (discovering user preferences) and exploitation (recommending highest-value items), combining NLP feature extraction with online reinforcement learning.

1.1 Success Metrics

Metric	Definition	Rationale
CTR@1	P(click | recommended)	Primary business metric
Avg Reward	Mean observed clicks across test set	Absolute performance
Avg Regret	Mean(optimal reward − selected reward)	Opportunity cost
Coverage	% of items with >0 recommendations	Diversity / fairness
Calibration	Category rec. % ≈ category prevalence %	Slice fairness
Hit@10	P(item in top-10 ranked candidates)	Ranking quality

nDCG@10	Normalized Discounted Cumulative Gain	Standard IR metric

1.2 Constraints

- Offline only - no online interaction with real users.
- Deterministic features - hashed BoW and one-hot category; no learned embeddings.
- Real-time latency - O(d) prediction per user (d = context dimension).
- Reproducibility - fixed seed = 42; full source code published under MIT License.
- Privacy - no PII; public dataset only.

2. DATASET, PREPROCESSING & SPLITS

2.1 Source & Governance

The **MIND Dataset (Microsoft News Dataset)** is a large-scale, publicly available news impression log published by Microsoft Research. License: research-only (non-commercial). The full dataset contains approximately 50,000 news articles and 6 million user behaviour records. A sampled subset of 30,000 events and 50,000 articles is used for all experiments.

- No PII stored — dataset is anonymized; no names, emails, or identifiable user data.
- Leakage prevention — train/validation split follows chronological order; no user data reuse across splits.
- Data retention — results and checkpoints stored in experiments/ (no raw behaviours committed).

2.2 Preprocessing

- Text: tokenise and lowercase (regex [a-z0-9]+); hash each token modulo 256; L1-normalise by token count → 256-dim.
- Category: hashed one-hot encode category and subcategory (32-dim each) → 64-dim vector.
- Entities: extract Wikidata entity IDs from JSON; average pre-trained 100-dim **embeddings; fall back to zero vector.**
- **User profile: mean of clicked-item embeddings; zero vector for cold-start users; updated offline daily.**
- **Context vector: [candidate_embedding ‖ user_profile ‖ (candidate − user_profile)] → 960-dim.**

2.3 Data Splits

Split	Size	Usage	Rationale
Train	**80% of events**	**Online RL learning**	**Maximize learning signal**
Validation	**20% of events**	**Offline evaluation**	**Unbiased test set**

Splits follow chronological order of impressions to mimic production deployment where future data is unseen at training time.

3. TRAINING DETAILS

3.1 Hyperparameters & Schedules

Parameter	Value	Justification
Text embedding dim	**256**	**Balanced vocabulary coverage**
Category dim	**32 per field**	**Lightweight, captures main categories**
Entity embedding dim	**100 (pre-trained)**	**Standard size, improves richness**
Total context dim	**960**	**3× (256+64+100) with interaction term**
LinUCB alpha (α)	**1.0**	**Moderate exploration (tunable)**
L2 regularization (LinUCB)	**1.0**	**Stable A matrix**
MLP hidden dim	**64**	**Lightweight but expressive**
MLP epochs	**3**	**Prevent overfitting**
MLP learning rate	**0.03**	**Standard SGD rate**
MLP batch size	**256**	**Memory/convergence tradeoff**
Train ratio	**0.8**	**Standard 80/20 split**
Random seed	**42**	**Full reproducibility**

3.2 Compute Environment

- Language: Python 3.8+, NumPy only — no GPU required.
- Feature build: ~30–40 s; Event loading: ~20–25 s.
- Training (LinUCB): ~5–10 s per epoch; Evaluation (all baselines): ~2–3 s; Total: ~60–80 s.

3.3 Reproducibility Steps

1. Clone: git clone https://github.com/francoiseee/recommender-sys-6intelsy.git
2. Install: pip install numpy>=1.21 (or conda env create -f environment.yml)

3. Download data: python data/get_data.py
4. Run full pipeline: python -m src.benchmark --data-dir data --max-events 30000 --max-news 50000 --text-dim 256 --category-dim 32 --seed 42
5. Results saved to results/tables/baseline_metrics.csv and results/figures/*.svg

4. MODELING

4.1 Architecture Overview

The system pipelines three components: (1) a feature engineering module that constructs 960-dim context vectors from news text, category, entity, and user history; (2) a set of four baseline recommenders; and (3) the proposed LinUCB contextual bandit.

4.2 Baeline Architectures

Random Selection: Uniform random arm selection. Establishes lower bound.

Popularity Ranking: Items scored by historical CTR = clicks/impressions; argmax selected. Strong non-personalised baseline.

Content-Similarity Cosine: score(i) = cos(candidate_embedding, user_profile). Simple personalisation without online learning.

Shallow MLP (NumPy): 2-layer network (960-dim → ReLU(64) → sigmoid(1)) trained via mini-batch SGD. Nonlinear learnable baseline.

4.3 Proposed Model: LinUCB Contextual Bandit

LinUCB treats each recommendation as an arm selection. The UCB score for candidate i at step t is:

$$UCB_i(t) = x_i^T \theta_t + \alpha \sqrt{(x_i^T A^{-1}_t x_i)}$$

The mean term estimates expected reward; the bonus encourages exploration. Parameters A (960×960) and b (960-dim) are updated via the Sherman-Morrison rank-1 formula. L2 regularisation (λ=1.0) ensures numerical stability.

4.4 NLP Components

- Hashed Bag-of-Words (256-dim): Fast, memory-efficient text representation. Limitation: loses word order and semantic structure.
- Hashed One-Hot Category (64-dim): Captures categorical structure explicitly; lightweight.
- Pre-trained Entity Embeddings (100-dim): Semantic richness from Wikidata entities; requires external embedding file.
- Interaction Term (candidate − user_profile): Explicitly encodes personalised relevance in the context vector.

4.5 Why LinUCB?

- Principled exploration-exploitation via upper confidence bounds — no manual ε schedule required.
- Sublinear regret bound: E[Regret_T] = O(d^{3/2} √T log T) (Abbasi-Yadkori et al., 2011).
- $O(d^2)$ per update with rank-1 Sherman-Morrison — no batch retraining needed.
- Linear reward model is interpretable and auditable; compatible with fairness constraints.

5. EVALUATION

5.1 Main Results

Model	**Events**	**CTR@1**	**Avg Reward**	**Avg Regret**	**Coverage**	**Hit@10**	
Random Baseline	6000	0.0100	0.0100	0.0200	87.3%	0.0850	
Popularity	6000	0.0120	0.0120	0.0180	45.2%	0.0920	
Content Similarity	6000	0.0110	0.0110	0.0190	72.1%	0.0890	
MLP DL Baseline	6000	0.0115	0.0115	0.0185	68.5%	0.0905	
LinUCB (α=1.0)	**6000**	**0.0135**	**0.0135**	**0.0165**	**78.4%**	**0.0945**	**BEST**

- LinUCB achieves +35% CTR improvement over the random baseline.
- LinUCB achieves +12.5% CTR improvement over the popularity baseline (best non-learning model).
- Coverage (78.4%) balances exploitation (popularity ≈ 45%) and exploration (random ≈ 87%).
- Hit@10 (0.0945) indicates improved ranking quality across top-10 candidates.

5.2 Learning Curves

LinUCB cumulative reward follows three phases: (1) steep initial rise (steps 1–1,000) as θ converges rapidly; (2) gradual plateau (steps 1,000–4,000) as exploitation

dominates; and (3) asymptotic behaviour (steps >4,000) with stable reward ≈ 0.0135 ± 0.001.

5.3 Ablation 1 — Exploration Coeffiecient (α)

α	CTR@1	Avg Regret	Coverage	Notes
0.0	0.0125	0.0175	45.3%	Pure exploitation; low diversity
1.0 (default)	**0.0135**	**0.0165**	**78.4%**	**Optimal balance**
2.0	0.0128	0.0172	89.1%	Over-exploration; reduced CTR
5.0	0.0110	0.0190	94.7%	Random-like behaviour

5.4 Ablation 2 — Context Feature Set

Feature Set	Dim	CTR@1	Coverage	Hit@10	Notes
Text only	256	0.0118	65.2%	0.0810	TF-IDF-like baseline
Text + Category	320	0.0132	72.5%	0.0875	+11.9% CTR
Text + Category + Entity	**420**	**0.0135**	**78.4%**	**0.0945**	**+1.3% CTR — best overall**
User-side only	420	0.0105	48.3%	—	Poor; needs candidate signal

Entity embeddings add +1.3% CTR. The interaction term (candidate − user_profile) is essential; using user-side features alone degrades CTR to 0.0105 and coverage to 48.3%.

5.5 Slice 1 — Performance by News Category

Category	Num Items	CTR@1	Hit@10	Avg Regret	Observations
US	2145	0.0142	0.0834	0.0158	Best performance
Sports	1203	0.0140	0.0821	0.0160	Strong; popular & consistent
Entertainment	892	0.0089	0.0634	0.0211	Underperforms (−37%)
Business	1456	0.0128	0.0782	0.0172	Good; stable
Science	687	0.0121	0.0715	0.0179	Lower volume
Health	543	0.0113	0.0681	0.0187	Niche; low engagement

Finding: Entertainment is 37% below average CTR. Hypothesised causes: shorter abstracts (fewer text features), fewer entity links, and potential dataset bias. Mitigation via category-specific feature scaling is identified as future work.

5.6 Slice 2 — Performance by user History Length

History Length	Users	Events	CTR@1	Std Dev	Notes
0 (cold-start)	234	347	0.0095	0.018	Weak signal — zero user profile
1–5 items	1823	2104	0.0118	0.016	Noisy history; few anchors
6–20 items	2891	3456	0.0138	0.012	Best performance
>20 items	1052	1093	0.0133	0.013	Some staleness; older preferences

Finding: Users with 6–20 prior interactions yield the best recommendations. Cold-start users (0 history) score 0.0095 CTR — a 31% penalty relative to the optimal group. A content-based fallback for cold-start is identified as priority future work.

5.7 Hard Failure Cases

Case 1 — **Entity Dominance**: A sports enthusiast received a tech article recommendation because its high entity link count caused entity embeddings to dominate over weak text-user alignment. Lesson: collaborative filtering signals can be

overwhelmed by entity richness when feature weights are not calibrated.

Case 2 — **Text Density Confounding**: A health-conscious reader received celebrity gossip because the gossip article's denser token count produced stronger text features. Lesson: hashed BoW can introduce noise; semantic embeddings (e.g., BERT) could resolve this.

6. ETHICS & POLICY

6.1 Risk Register

Risk	Likelihood	Impact	Mitigation
Filter Bubbles / Echo Chambers	High	High	α=1.0 forces ~15% random exploration; coverage across categories measured; fairness constraints planned
Privacy & Consent	Low	High	Public dataset only; no PII; anonymized; clear data governance documentation
Popularity Bias / Unfair Item Exposure	Medium	Medium	Diversity metrics (coverage=78%); per-category exposure analysis; re-ranking with fairness as future work
Simulation Gap	High	Medium	Clear disclaimers in README and Model Card; click ≠ satisfaction; offline-only caveat
Misuse for Manipulation	Low	High	Academic-only use; no deployment pipeline; clear disclaimer in README and ethics statement
Cold-start Harm	Medium	Low	Content-based fallback recommended; random exploration mitigates; hybrid approach in future

6.2 Fairness Checks Implemented

Category Exposure Fairness: CTR computed per category; Entertainment underperformance flagged at −37% below average.

User Segment Fairness: Performance analysed by history length; cold-start users scored 11.9% lower — cold-start penalty documented.

Popularity Bias Measurement: Coverage = 78.4% (vs. random = 87.3%); far less concentrated than pure exploitation (45%).

6.3 Intended Use & Limitations

- Intended for: personalised news recommendation research; offline experimentation; academic teaching of contextual bandits + NLP; benchmarking.
- NOT intended for: real user deployment without fairness guardrails; high-stakes decisions; systems targeting vulnerable populations; engagement maximisation at cost of user wellbeing.

6.4 Transparency & Reproducibility

- All model decisions and hyperparameters documented in this report and experiments/configs/.
- Fixed seed (seed=42) throughout all pipeline stages.
- Code is open-source under the MIT License; failure cases and slice analyses are documented.

7. MODEL CARD

Field	Value
Name	LinUCB Contextual Bandit News Recommender
Version	1.0 (Final, Week 3)
Type	Contextual Multi-Armed Bandit (LinUCB)
Input	User history (news IDs) + candidate items (title, abstract, category, entities)
Output	Top-1 item recommendation + UCB confidence score
Metrics	CTR@1, Avg Reward, Avg Regret, Coverage, Calibration
Data	MIND Dataset (public, anonymized, research-only)
Training Set	~24,000 impressions (80% of 30k loaded)
Validation Set	~6,000 impressions (20%)
Training Time	~5–10 seconds per full epoch (NumPy only)

Inference Time	O(d) ≈ 1–5 ms per impression (d=960)
Model Size	~30 MB uncompressed (A: 960×960 matrix; b, theta: 960-dim vectors)
Caveats	Offline evaluation only; short-term click proxy; cold-start weakness; entertainment category bias

7.1 Deployment Guidance & Disclaimers

- Offline evaluation only — simulation gap exists; click labels may not reflect live satisfaction.
- Cold-start users require a content-based fallback before production deployment.
- Entertainment category bias requires investigation and fairness constraints before deployment.
- CTR is a short-term proxy; dwell time and return rate should be incorporated in live evaluation.
- System must not be repurposed to manipulate users, surveil individuals, or make consequential decisions (health, legal, finance, hiring).
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8. HOW TO RUN

8.1 Environment Setup

Option A — pip: pip install numpy>=1.21 (Python 3.8+, OS: Linux / macOS / Windows)

Option B — conda: conda env create -f environment.yml && conda activate recommender-sys

8.2 One-Command Reproduction

```
python -m src.benchmark --data-dir data --max-events 30000 --max-news 50000 --seed 42
```

Output: results/tables/baseline_metrics.csv, results/figures/linucb_learning_curve.svg, results/figures/baseline_ctr_comparison.svg, experiments/checkpoints/linucb_run_final.json.

8.3 Demon Instructions

1. Clone: git clone https://github.com/francoiseee/recommender-sys-6intelsy.git
2. Set up environment (Section 8.1).
3. Download MIND data: python data/get_data.py
4. Run the one-command benchmark from Section 8.2.
5. Inspect results/tables/baseline_metrics.csv for per-model performance numbers.
6. View results/figures/linucb_learning_curve.svg for training dynamics visualisation.

9. LIMITATIONS & FUTURE WORK

9.1 Current Limitations

1. Offline evaluation only — simulation gap; click ≠ satisfaction; no live A/B test conducted.
2. Hashed features — word order and semantics lost; fixed hash collisions; no parameterisation.
3. Cold-start — zero-history users receive marginal recommendations; no fallback implemented.
4. Temporal drift — by-order split used; concept drift and seasonal trends not explicitly handled.
5. Short-term reward — clickbait can inflate CTR; dwell time and return rate not captured.
6. Category imbalance — Entertainment −37% CTR; root cause unclear.
7. Scalability — $O(d^2)$ per update; 960-dim is manageable but larger contexts are challenging.

9.2 Recommended Future Work

Short-term (1–2 weeks): Online evaluation, cold-start content-based ranker, temporal re-split, multi-seed confidence intervals.

Medium-term (1–2 months): BERT/RoBERTa text embeddings, multi-objective optimisation (CTR + diversity + fairness), category fairness constraints, user clustering.

Long-term (3–6 months): Causal inference for long-term value, Thompson Sampling / LinTS variants, fairness-aware bandit formulations, hybrid content + collaborative + contextual recommenders.

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